

Introduction to Biophysics II: Quantum physics & biology

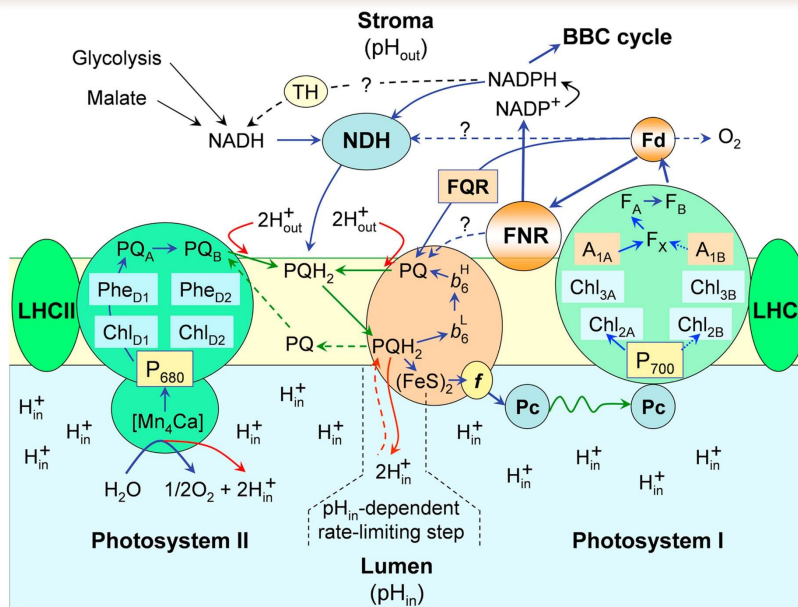
Lecture 14: Electron transfer in biology

Reading:

- Quantum Effects in Biology, chapter 9
- Bolton et al.; Electron Transfer in Inorganic, Organic, and Biological Systems. Advances in Chemistry; American Chemical Society: Washington, DC, 1991
- Newton MD, Chem. Rev.91, 767, 1991
- Marcus RA, Sutin N. BBA 811 (1985) 265
- Marcus RA, JCP 24 (1956) 966
- C.C. Page, C.C. Moser, X.X. Chen, P.L. Dutton, Natural engineering principles of electron tunneling in biological oxidation-reduction, Nature 402 (1999) 47–52

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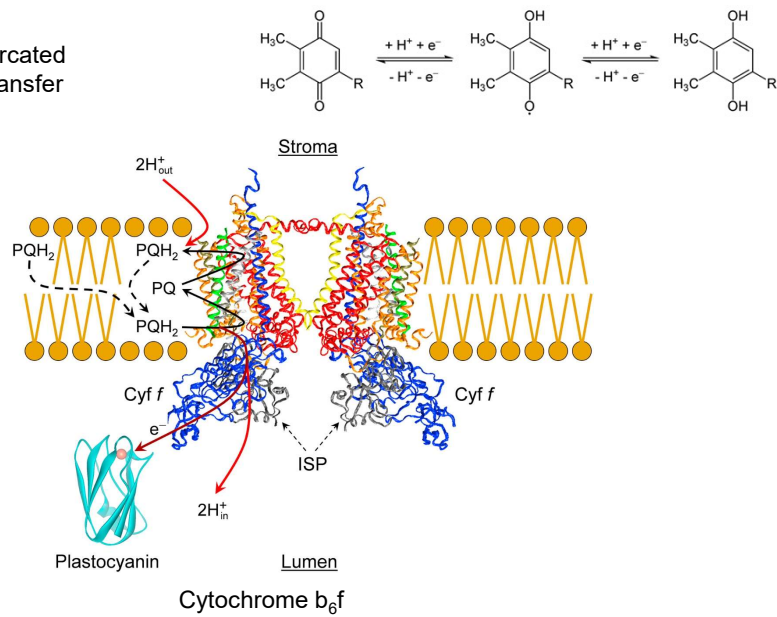
Charge transfer: Photosynthesis



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Electron transfer: example

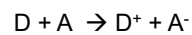
Cyclic bifurcated
electron transfer



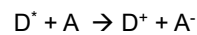
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Electron transfer: scenarios

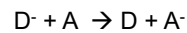
Transfer of electron from donor to acceptor:



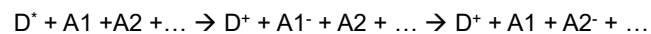
Transfer of electron from excited donor to neutral acceptor:



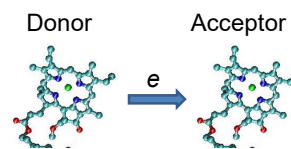
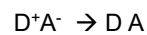
Transfer of electron from reduced donor to neutral acceptor:



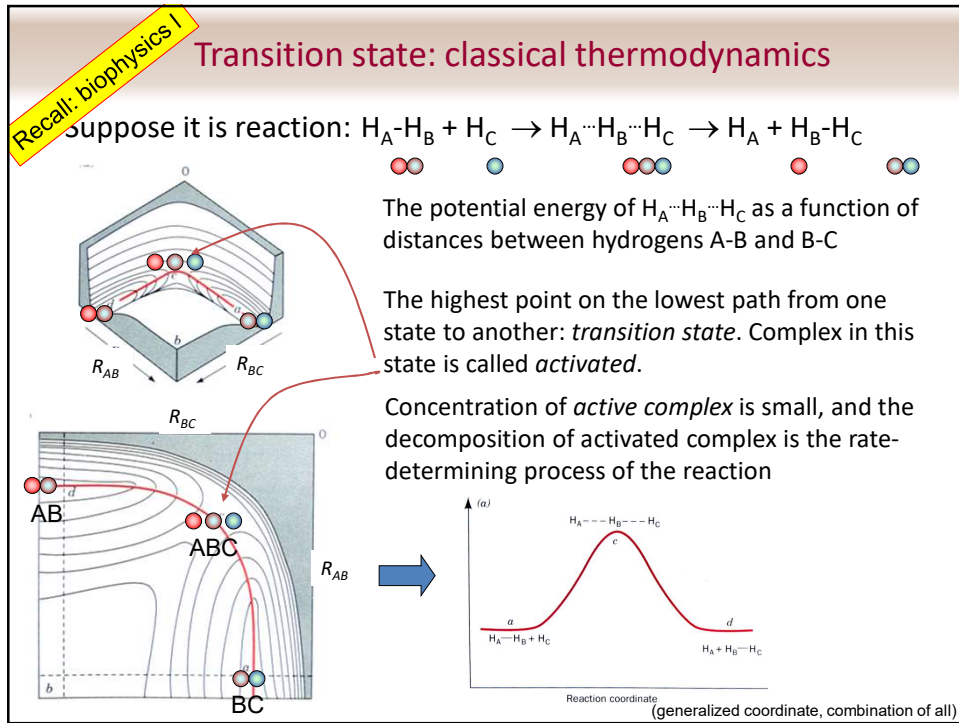
Transfer through a chain of cofactors:



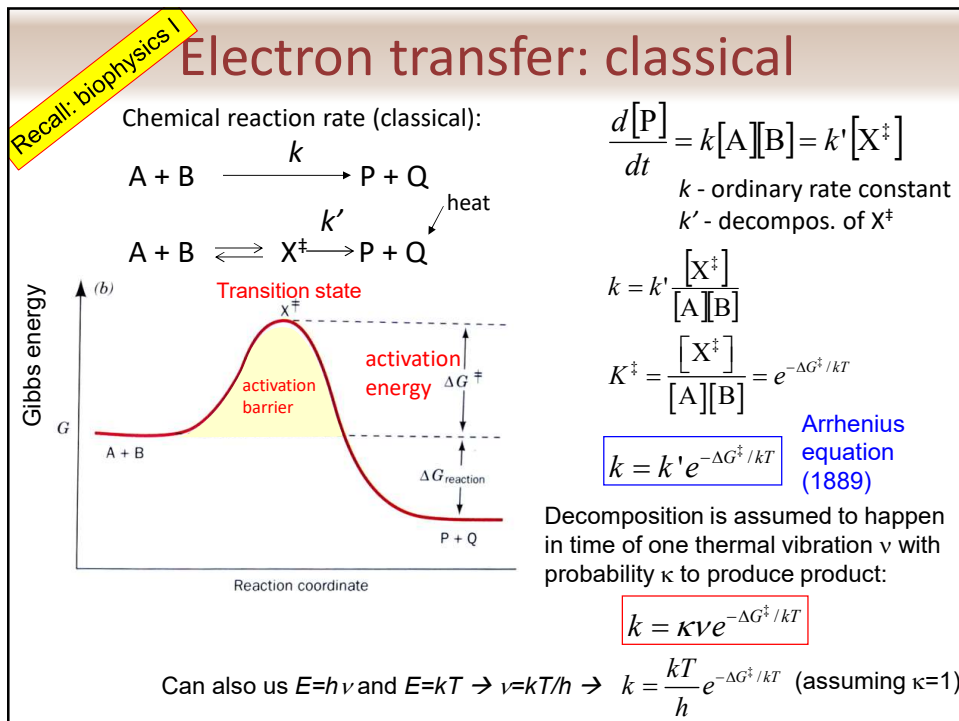
Recombination:



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Electron transfer: Marcus theory

Marcus' theory, developed in 1956-1965, Nobel prize 1992
classical approach (non-adiabatic)
 Transfer of an electron from donor to acceptor :

$$D + A \rightarrow D^+ + A^-$$

First, we assume that the potential energy is represented by parabolas (harmonic oscillator):

G_D – Potential (Gibbs) energy of state when an electron is on the donor

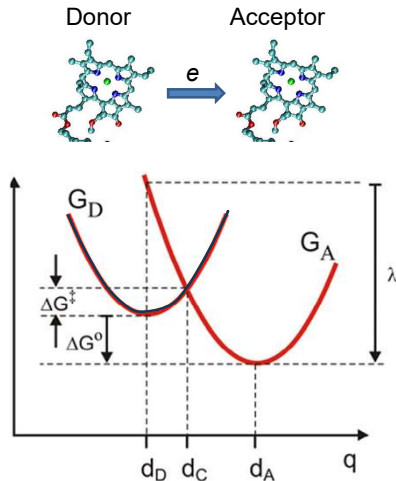
G_A – Potential energy when an electron is on the acceptor

ΔG_0 – difference in energies of initial and final states (relaxed)

q – generalized reaction coordinate

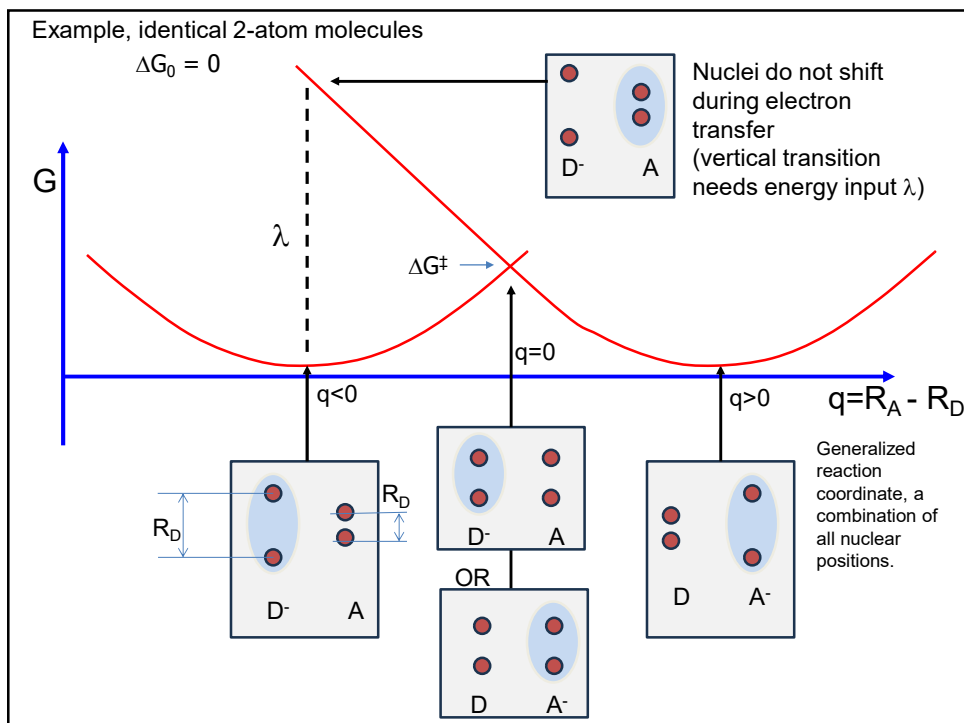
$\Delta G^\ddagger$ - activation barrier

λ - reorganizational energy



Marcus theory: pages are adopted from:
 Download Center <https://chem.libretexts.org/@go/page/134526> (accessed Mar 14, 2021).

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Electron transfer...

Marcus: classical approach (non-adiabatic)

Parabolas (harmonic oscillator):

Following Arrhenius equation for transition state theory:

$$k = \kappa \nu e^{-\Delta G^\ddagger / kT}$$

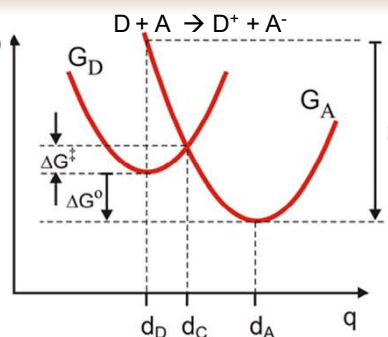
The probability κ is here the probability for an electron to transfer/tunnel from donor to acceptor, denoted as k_{el} :

$$k = k_{el} \nu e^{-\Delta G^\ddagger / kT}$$

ν is the frequency of passage (nuclear motion) of the transition state at $q=d_C$.

In classical treatment electron transfer k_{el} is taken to be unity (as κ) – not a valid approach for electron transfer!

Need to know $\Delta G^\ddagger$



Note: $\Delta G^0 < 0$

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Electron transfer...

Marcus: classical approach (non-adiabatic)

Parabolas (harmonic oscillator):

$$G_D(q) = \frac{1}{2} m \omega_0^2 (q - d_D)^2$$

$$G_A(q) = \frac{1}{2} m \omega_0^2 (q - d_A)^2 + \Delta G^0$$

The barrier height at the crossing point d_C :

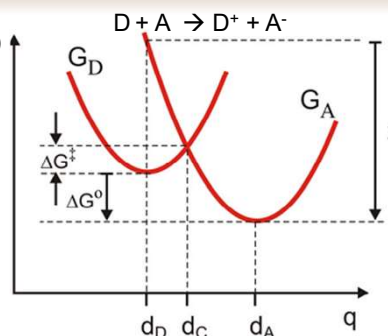
$$G_D(d_C) = G_A(d_C)$$

$$\frac{1}{2} m \omega_0^2 (d_C - d_D)^2 = \Delta G^0 + \frac{1}{2} m \omega_0^2 (d_C - d_A)^2$$

$$d_C = \frac{\Delta G^0}{m \omega_0^2} \left(\frac{1}{d_A - d_D} \right) + \frac{d_A + d_D}{2}$$

Note that $\rightarrow \lambda = G_A(d_D) - G_A(d_A) = \frac{1}{2} m \omega_0^2 (d_D - d_A)^2$
And we get:

$$d_C = \frac{\Delta G^0}{2\lambda} (d_A - d_D) + \frac{d_A + d_D}{2}$$



Note: $\Delta G^0 < 0$

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Electron transfer...

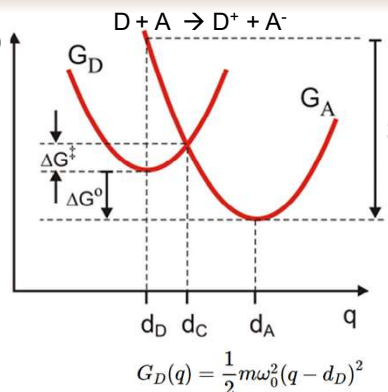
Marcus: classical approach (non-adiabatic)

$$\text{Have: } d_C = \frac{\Delta G^\circ}{2\lambda}(d_A - d_D) + \frac{d_A + d_D}{2}$$

$$\lambda = \frac{1}{2}m\omega_0^2(d_D - d_A)^2$$

And the barrier height:

$$\begin{aligned}\Delta G^\ddagger &= G_D(d_C) - G_D(d_D) \\ &= \frac{1}{2}m\omega_0^2(d_C - d_D)^2 \\ &= \frac{1}{4\lambda}[\Delta G^\circ + \lambda]^2\end{aligned}$$



Plug this into Arrhenius equation: $k = k_{el}\nu e^{-\Delta G^\ddagger/kT}$

And we get

$$k_{ET} = k_{el}\nu \exp\left(-\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}\right)$$

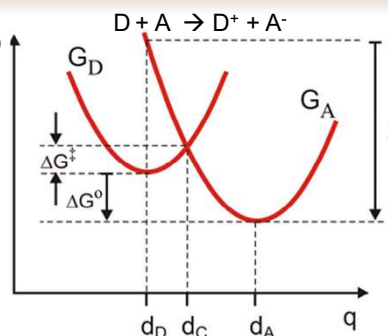
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Electron transfer...

Marcus: classical approach (non-adiabatic)

$$k_{ET} = k_{el}\nu \exp\left(-\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}\right)$$

- Electron transfer rate depends exponentially on temperature
- Electron transfer maximizes when $\Delta G^\circ = -\lambda$



$k_{el} = ? \rightarrow$ electron transfer involves *tunneling* from one location to another.

Note: Distance between donor and acceptor, R_{DA} , is not equal to $d_A - d_D$!!!

q is the "reaction coordinate" that involves a complex trajectory in a multidimensional space comprised of coordinates of *all atoms* in the system that shift as the result of electron transfer

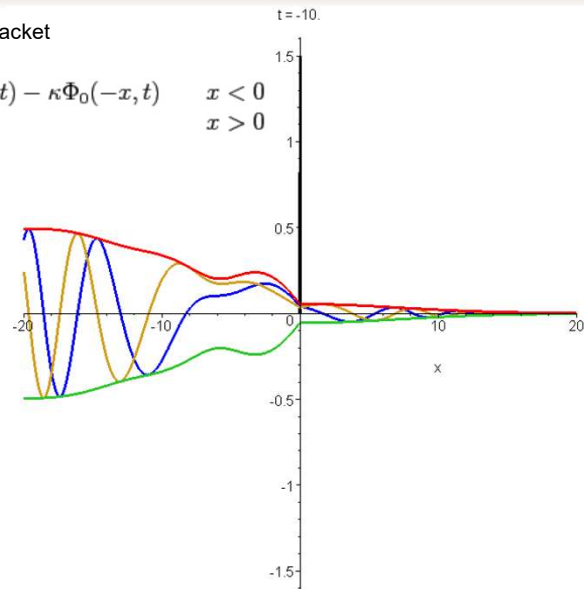
R is the *actual* distance between the donor and acceptor to be tunneled by the electron

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Tunneling

Free space Gaussian wavepacket

$$\Phi(x, t) = \begin{cases} (-\partial_x + \kappa)\Phi_0(x, t) - \kappa\Phi_0(-x, t) & x < 0 \\ -\partial_x\Phi_0(x, t) & x > 0 \end{cases}$$

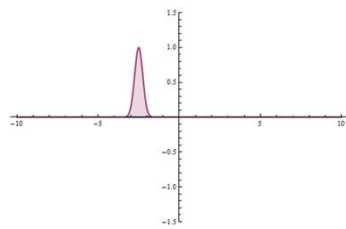


<https://www.victoria.ac.nz/scps/about/staff/johnlekner>

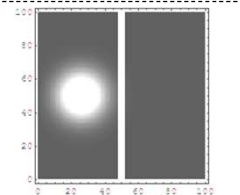
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Tunneling

Narrow barrier



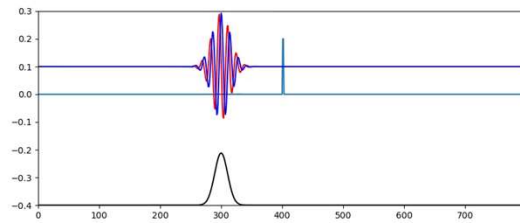
Electron tunneling



http://en.wikipedia.org/wiki/Quantum_tunnelling

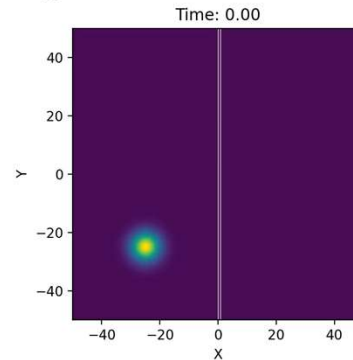
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Tunneling



Python simulations:

<https://savikhin.github.io/Python/>



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Tunneling: square barrier

Tunneling through a square barrier.

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)$$

$$V(x) = \begin{cases} 0, & \text{within free space} \\ V_0, & \text{within barrier} \end{cases}$$

Solve SE: $\hat{H}\psi = E\psi \Rightarrow \frac{\partial^2 \psi}{\partial x^2} = -2m\hbar^{-2}(E - V)$

$$\frac{\partial^2 \psi_{1,2}}{\partial x^2} = \frac{-2mE\hbar^{-2}}{\uparrow < 0} \psi_{1,2}, \text{ within free space}$$

$$\frac{\partial^2 \psi_3}{\partial x^2} = \frac{-2mE\hbar^{-2}(E - V_0)}{\uparrow > 0} \psi_3, \text{ within barrier}$$

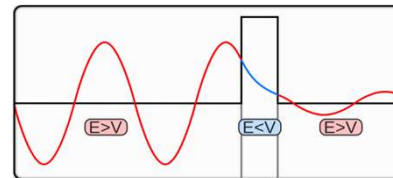
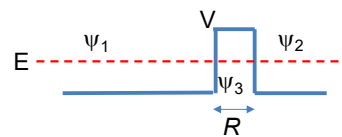
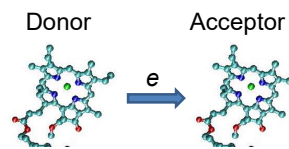
$$\text{Recall: } \frac{\partial^2 e^{kx}}{\partial x^2} = k^2 e^{kx}$$

$$\psi_1 = Ae^{ikx} + Be^{-ikx} \quad \text{Harmonic wave}$$

$$\psi_2 = Ce^{ikx} + De^{-ikx} \quad k = \sqrt{2mE} / \hbar$$

$$\psi_3 = Fe^{\kappa x} + Ge^{-\kappa x} \quad \kappa = \sqrt{2m(V_0 - E)} / \hbar$$

Exponential decay



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Tunneling: square barrier

Tunneling through square barrier.

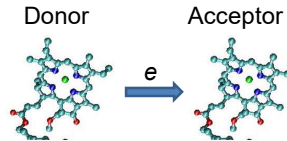
$$\psi_1 = Ae^{ikx} + Be^{-ikx}$$

$$\psi_2 = Ce^{ikx} + De^{-ikx}$$

$$\psi_3 = Fe^{\kappa x} + Ge^{-\kappa x}$$

$$k = \sqrt{2mE} / \hbar$$

$$\kappa = \sqrt{2m(V_0 - E)} / \hbar$$



1. Boundary conditions: functions and their first derivatives of x must be continuous (d/dx^2 must exist!).
2. We are interested in coefficients A (moving right) and C (moving right)
3. Transmission:

$$T = \left| \frac{C}{A} \right|^2 = \left| \frac{4ikme^{-ikR-mR}}{(k+ik)^2 - (k-ik)^2 e^{-2\kappa R}} \right|^2 = \frac{1}{1 + V_0^2 \sinh^2(\kappa R) / [4E(V_0 - E)]}$$

For high and/or wide barrier ($\kappa R \gg 1$): $\sinh x = \frac{e^x - e^{-x}}{2}$

$$T \approx \frac{16E(V_0 - E)}{V_0^2} \exp(-2\kappa R) = \frac{16E(V_0 - E)}{V_0^2} \exp\left(-\frac{2R}{\hbar} \sqrt{2m(V_0 - E)}\right)$$

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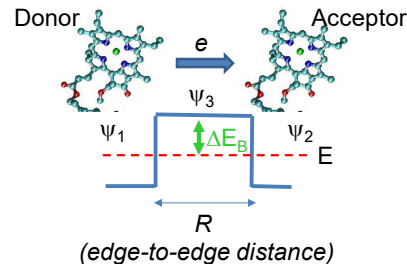
Tunneling: square barrier

Tunneling through high/wide square barrier:

$$T = \frac{16E(V_0 - E)}{V_0^2} \exp\left(-\frac{2R}{\hbar} \sqrt{2m(V_0 - E)}\right)$$

Denote $\Delta E_B = V_0 - E$, rewrite:

$$T = \frac{16E\Delta E_B}{V_0^2} \exp\left(-\frac{2R}{\hbar} \sqrt{2m\Delta E_B}\right)$$



Tunneling depends exponentially on the (distance) x (square root of barrier height).
Exponential term dominates transmission rate at large distances:

$$T \propto \exp\left(-\frac{2R}{\hbar} \sqrt{2m\Delta E_B}\right) \quad (\text{where } \Delta E_B \text{ is average barrier height for uneven barriers})$$

And we can rewrite $k_{ET} = k_e \nu \exp\left(-\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}\right)$

$$k_{ET} \propto \exp\left(-\frac{2R}{\hbar} \sqrt{2m\Delta E_B}\right) \exp\left(-\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}\right)$$

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Activation barrier + tunneling

Electron transfer:

$$k_{ET} \propto \exp\left(-\frac{2R}{\hbar}\sqrt{2m\Delta E_b}\right) \exp\left(-\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}\right)$$

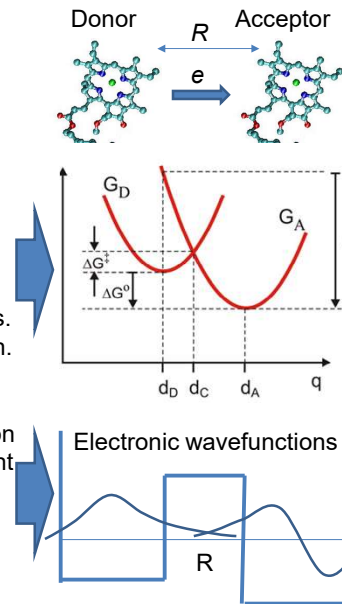
$$\exp\left(-\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}\right)$$

The second term is a requirement that the energies of the initial and final states match!

This is the probability that the electronic + vibrational energy of the donor and acceptor cross. $\Delta G^\ddagger$ - is the activation barrier for a nuclear reaction. q is the reaction coordinate

$$\exp\left(-\frac{2R}{\hbar}\sqrt{2m\Delta E_b}\right)$$

The first term is due to the electron tunneling, ΔE_b is the barrier height for electron tunneling!
 R is the actual distance along the line connecting donor and acceptor.



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Electron transfer...

Sometimes it is expressed in terms similar to quantum mechanical approach:

$$k_{ET} = \frac{2\pi}{\hbar} H_{DA}^2 \rho_{FC}^{clas}$$

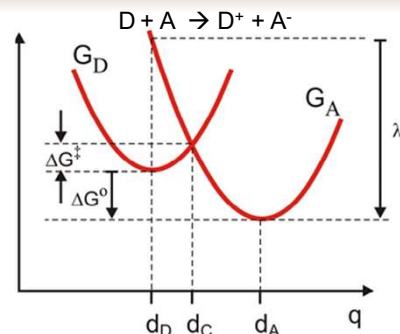
H_{DA}^2 Tunneling matrix element,

$$H_{DA} \propto \sqrt{T} \propto \exp\left[-\left(\frac{\sqrt{2m_e\Delta E_b}}{\hbar}\right)R_{DA}\right]$$

ρ_{FC}^{clas} Classical Boltzmann activation factor ("classical Franck-Condon factor")

$$\rho_{FC}^{clas} = \frac{1}{\sqrt{2\pi\sigma_{\Delta G}^2}} \exp\left(-\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}\right)$$

$$\ln k_{ET} = const - \left(\frac{2\sqrt{2m_e\Delta G_{average}^\ddagger}}{\hbar}\right)R_{DA} - \frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}$$



$\sigma_{\Delta G}^2$ - Variance in the energy gap $G_D - G_A$:

$$\sigma_{\Delta G}^2 = 2\lambda kT$$

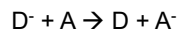
From: Quantum effects in biology

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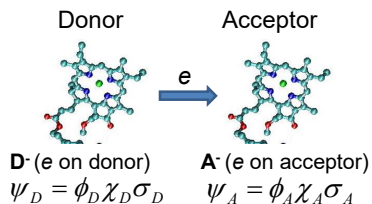
Electron transfer

Quantum-mechanical approach

Transfer of electron from reduced donor to neutral acceptor:



Consider wave function of electron on donor (initial state) and on acceptor (final):
(Born-Oppenheimer approximation)



Construct superposition state for the electron:

$$\Psi = C_D \phi_D \chi_D \sigma_D + C_A \phi_A \chi_A \sigma_A$$

ϕ - spatial wave function
 χ - nuclear wave function
 σ - spin

Initial state: $C_D=1, C_A=0$.

Need to solve time-dependent Schrödinger equation to find $C_A(t)$: $\hat{H}\Psi = i\hbar \frac{\partial \Psi}{\partial t}$

The solution is mathematically similar to Förster/Dexter energy transfer:

Recall
Lecture 6

$$C_2(\tau) = H_{21} (1 - \exp[i(E_2 - E_1)\tau/\hbar]) / (E_2 - E_1)$$

$$C_A \quad \langle \psi_2 | \tilde{H}_{21} | \psi_1 \rangle \leftarrow \langle \phi_A \chi_A \sigma_A | H_{AD} | \phi_D \chi_D \sigma_D \rangle$$

$$\begin{matrix} E_1 = E_D \\ E_2 = E_A \end{matrix}$$

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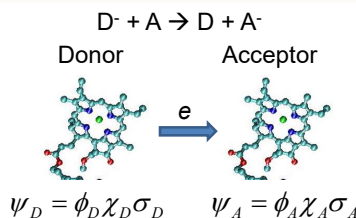
Electron transfer

Quantum-mechanical approach

$$\Psi = C_D \phi_D \chi_D \sigma_D + C_A \phi_A \chi_A \sigma_A$$

Initial state: $C_D=1, C_A=0$:

$$\langle \phi_A \chi_A \sigma_A | H_{AD} | \phi_D \chi_D \sigma_D \rangle$$



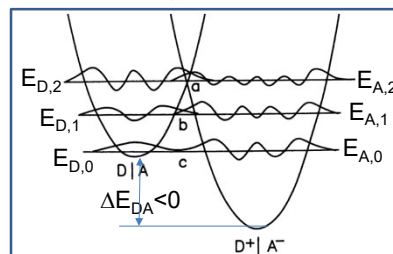
Solution:

$$k_{rt} = \frac{2\pi}{\hbar} |H_{21(el)}|^2 \sum_{n,m} \left[\frac{\exp(-E_{D,n}/kT)}{Z_1} \left| \langle \chi_{A,m} | \chi_{D,n} \rangle \right|^2 \times \delta(E_{A,m} - E_{D,n} + \Delta E_{DA}) \right] \times \langle \sigma_A | \sigma_D \rangle$$

Energy in vib. state m of acceptor
Energy in vib. state n of donor

Recall Lecture 7:

$$k_{rt} = \frac{2\pi}{\hbar} |H_{21(el)}|^2 \sum_{n,m,u,w} \left[\frac{\exp(-E_{n(1)}/kT)}{Z_1} \left| \langle \chi_{1m} | \chi_{1n} \rangle \right|^2 \times \frac{\exp(-E_{u(2)}/kT)}{Z_2} \left| \langle \chi_{2w} | \chi_{2u} \rangle \right|^2 \times \delta(\Delta E_{1nm} - \Delta E_{2uw}) \right]$$



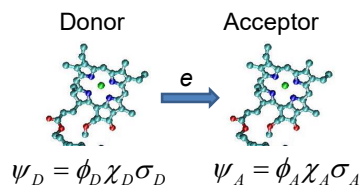
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Electron transfer

Quantum-mechanical approach

Solution:

$$k_{rt} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 \times \sum_{n,m} \left[\frac{\exp(-E_{D,n}/kT)}{Z_1} |\langle \chi_{A,m} | \chi_{D,n} \rangle|^2 \right] \times \left[\times \delta(E_{A,m} - E_{D,n} + \Delta E_{DA}) \right] \times \langle \sigma_A | \sigma_D \rangle$$



1. Interaction

Recall: Born-Oppenheimer approximation, electron wavefunction separates from nuclear:

$$\begin{aligned} \langle \phi_A \chi_A \sigma_A | \tilde{H}_{AD} | \phi_D \chi_D \sigma_D \rangle &= \langle \phi_A(\vec{r}) \chi_A(\vec{R}) \sigma_A(s) | \tilde{H}_{AD} | \phi_D(\vec{r}) \chi_D(\vec{R}) \sigma_D(s) \rangle \\ &= \langle \phi_A | \tilde{H}_{AD} | \phi_D \rangle \langle \chi_A | \chi_D \rangle \langle \sigma_A | \sigma_D \rangle \end{aligned}$$

$$H_{AD(el)} \equiv \langle \phi_A | \tilde{H}_{AD} | \phi_D \rangle$$

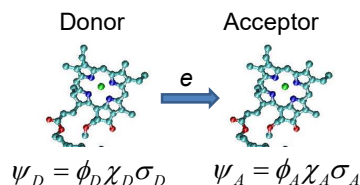
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Electron transfer

Quantum-mechanical approach

1. Electron interaction:

$$H_{AD(el)} \equiv \langle \phi_A | \tilde{H}_{AD} | \phi_D \rangle$$

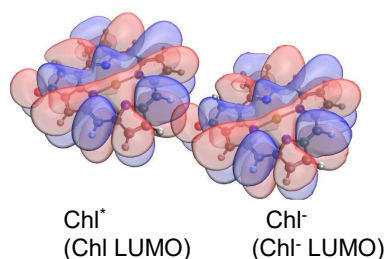


In the simplest form this is just electron tunneling:

$$H_{AD} \propto \sqrt{T} \propto \exp \left[- \left(\frac{\sqrt{2m_e \Delta E_b}}{\hbar} \right) R_{DA} \right]$$

Since wavefunctions decay far from molecule exponentially, this term gives exponential dependence on the distance.

Example:



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Electron transfer

Quantum-mechanical approach

1. Electron interaction:

$$H_{AD(el)} \equiv \langle \phi_A | \tilde{H} | \phi_D \rangle$$

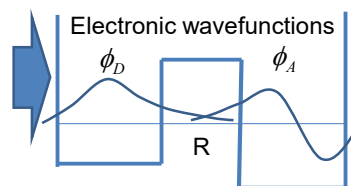
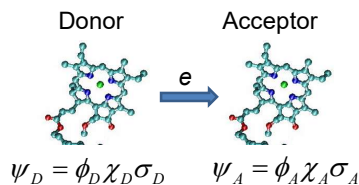
Alternative simplified approach:
what is the probability of finding an electron
with wave function ϕ_D on the acceptor?

The probability would be:

$$H_{DA(el)} \propto \langle \phi_A | \phi_D \rangle$$

(Fourier trick)

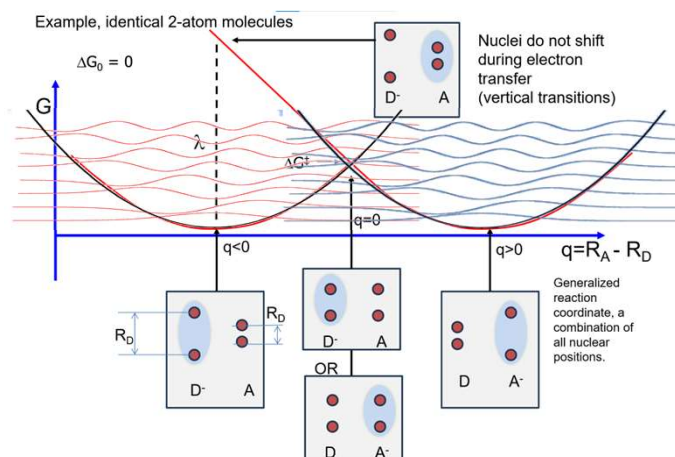
Overlap of electronic wavefunctions on donor and acceptor.



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Electron transfer

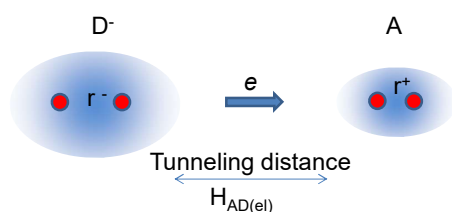
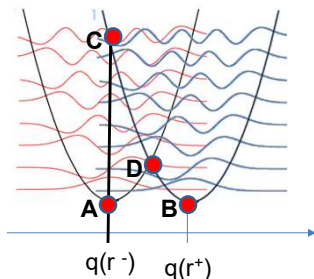
Recall, earlier: classical barrier (Marcus)



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Electron transfer

The meaning of the nuclear term: homodimer of diatomic molecules



$$k_{et} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 \times \sum_{n,m} \left[\frac{\exp(-E_{D,n}/kT)}{Z_1} |\langle \chi_{A,m} | \chi_{D,n} \rangle|^2 \right] \times \left[\times \delta(E_{A,m} - E_{D,n} + \Delta E_{DA}) \right] \times \langle \sigma_A | \sigma_D \rangle$$

Left parabola: Nuclear energy dependence on distance between atoms with electron on donor
Right parabola: Nuclear energy dependence on distance between two atoms with missing electron (acceptor)

Points:

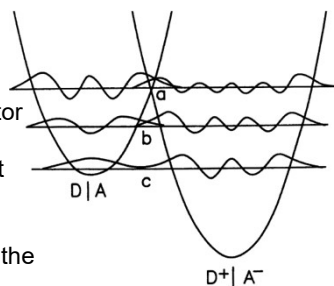
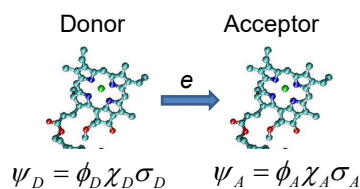
A: equilibrium distance with electron
B: equilibrium distance without electron
C: energy with electron on acceptor at equilibrium distance of molecule with electron; $E_C - E_A = \lambda$ (reorg. energy)
D: $r^- = r^+$ - electron on acceptor or donor will have the same energy

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Electron transfer

Quantum-mechanical approach

$$k_{et} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 \times \sum_{n,m} \left[\frac{\exp(-E_{D,n}/kT)}{Z_1} |\langle \chi_{A,m} | \chi_{D,n} \rangle|^2 \right] \times \left[\times \delta(E_{A,m} - E_{D,n} + \Delta E_{DA}) \right] \times \langle \sigma_A | \sigma_D \rangle$$



2. Nuclear term:

- involves the overlap of nuclear wave functions
- Vibrational wave functions of the donor and acceptor must overlap along the reaction coordinate.
- these are vibrations of the Donor and Acceptor, not overlap between the ground and excited state of a single molecule (not like in energy transfer).
- Unlike classical physics, the transfer is possible at the lowest temperatures due to *tunneling*.
- FC's could be determined experimentally from the absorption spectra of donor and acceptor

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Electron transfer

Quantum-mechanical approach

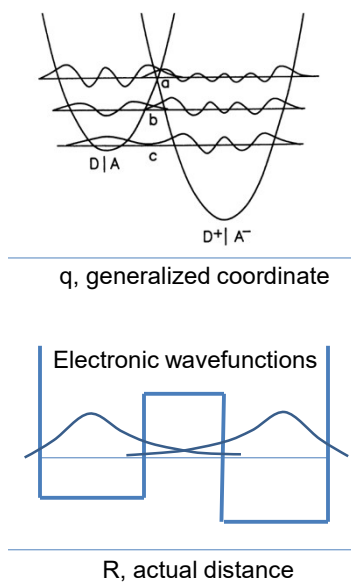
Overlap of nuclear

$$\langle \chi_{A,m} | \chi_{D,n} \rangle$$

and

electronic wavefunctions

$$H_{DA(el)} \propto \langle \phi_A | \phi_D \rangle$$



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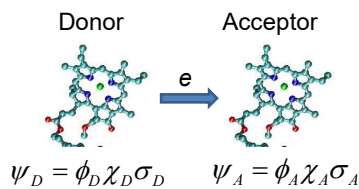
Electron transfer

Quantum-mechanical approach

$$k_{et} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 \times \sum_{n,m} \left[\frac{\exp(-E_{D,n}/kT)}{Z_1} |\langle \chi_{A,m} | \chi_{D,n} \rangle|^2 \times \delta(E_{A,m} - E_{D,n} + \Delta E_{DA}) \right] \times \langle \sigma_A | \sigma_D \rangle$$

3. Spin term

- Spin must be conserved!



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Electron transfer

Quantum-mechanical approach:

Nonadiabatic (or diabatic):

There are no mixed states, electron is on either the donor or the acceptor.

Electron transfer can occur:

a) Tunneling at the transition state:

The reactant (DA) and product (D^+A^-) have the same nuclear configuration, even when H_{AD} electron transfer can occur with high probability. However, transfer is temperature-dependent (even though electron tunneling is not)

b) Activated nuclear tunneling: nuclear surfaces are close enough, and nuclear wavefunctions overlap. An electron can tunnel at energies below the transition point (tunnel in terms of vibrational wf, not electron tunneling between molecules)

c) Temperature-independent nuclear tunneling: Even at the lowest temperature, there is overlap of nuclear wave functions.

For example, electron transfer in PS I RC at liquid helium temps has been shown experimentally to proceed at rates comparable to room temperature

